Final Architecture and Implementation Notes

Greater Darwin Property-Price Forecasting, Group Dan1-Theme2 (PRT661)
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Final system architecture

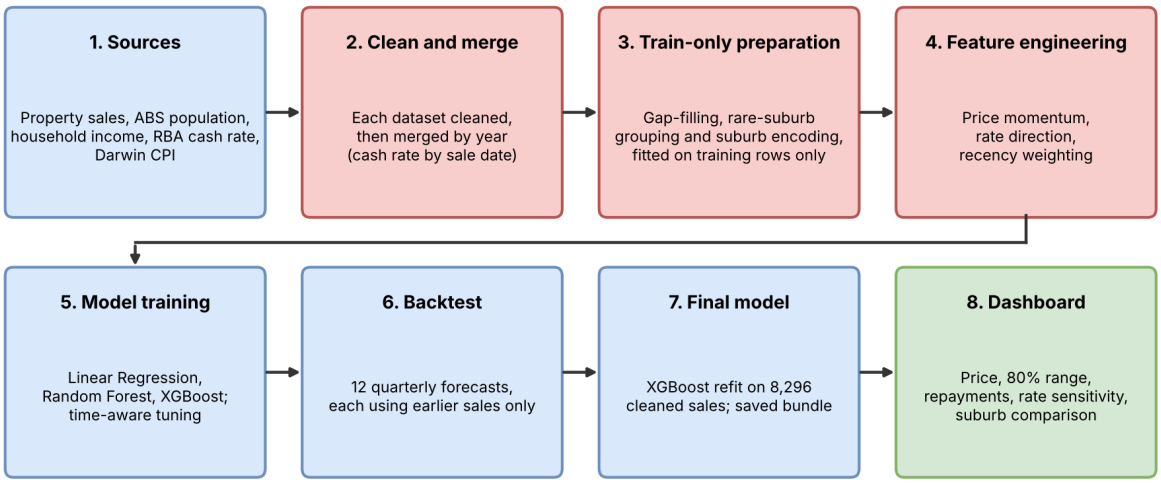


Figure: Final system architecture (eight stages)

The pipeline has eight stages. Sources: Homely property sales, ABS population, NT household income, RBA cash rate and Darwin CPI. Clean and merge: each dataset is cleaned, then merged by year (the cash rate by sale date). Train-only preparation: gap-filling, rare-suburb grouping and suburb encoding are fitted on training rows only. Feature engineering: price momentum, rate direction and recency weighting. Model training: Linear Regression, Random Forest and XGBoost with time-aware tuning. Backtest: twelve quarterly forecasts, each using only earlier sales. Final model: XGBoost refit on 8,296 cleaned sales. Dashboard: price, 80% range, repayments, rate sensitivity and suburb comparison.

Repository structure

- **code/main.ipynb**: the pipeline notebook (cleaning, preparation, models, backtest, final model)
- **code/plots.ipynb**: the plotting notebook
- **code/app.py**: the Streamlit dashboard
- **docs/**: architecture document and diagrams
- **data/**: source files and saved outputs
- **requirements.txt, README.md**: setup and run instructions

Working practice

- The main branch is protected. Every change goes through a pull request and is reviewed before it is merged.
- Work is tracked in Jira (project PDSPDT), with a ticket for each task.

How the pipeline works

The notebook runs the eight stages in order. The shared prepare_split function is the centre of it. It takes a cutoff date, splits the data in time, and fits gap-filling, rare-suburb grouping and suburb encoding on the training rows only. The baseline models, the robustness checks, the policy tests and the backtest all

call the same function, so every result uses the same leak-safe preparation.

Model bundle

After the backtest selected XGBoost, the model was refit on all 8,296 cleaned sales and saved with the encodings it needs. The dashboard loads this one bundle, so what a buyer sees always comes from the same fitted model.

Dashboard

The dashboard is a Streamlit app deployed at https://data-science-practice.streamlit.app/. It follows a five-screen buyer journey: choose a property, see the price and range, see repayments, test lender margins against the 30% stress line, and compare suburbs. Its five tabs are Overview, Model performance, Diagnosis to Fix, Predict and Affordability, and Limitations. Stamp duty, insurance and fees are not included in the repayment figures, and the dashboard says so.

Status and known limits

- The pipeline, the saved model and the deployed dashboard are working.
- The dashboard market context (latest sale, cash rate and CPI) is fixed at the last training sale, December 2025. Refreshing it needs a scheduled data update that is not built yet.
- Some location coordinates in the data are wrong. Correcting them is not finished.
- Both items are recorded as limitations and as future improvements in the final report.